

Probabilistic Spectral Unmixing for Materials Characterization

A Tutorial on Partial Knowledge of Composition and Spectra

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Abstract

Spectral unmixing arises whenever the probed volume contains several phases, oxidation states, coordination environments, or other material states with distinct spectral responses. In most materials experiments this is not a blind factorization problem. Composition is constrained by synthesis records or an independent assay, while the component spectra are constrained by reference measurements, physical line-shape models, simulations, or spectral libraries. The problem is to combine these sources without treating uncertain information as exact. This tutorial develops the Stage-1 formulation in which composition and mixture fraction are identified and the endmember spectra are globally shared. It starts from representative materials experiments, reviews the related literature, derives the exact limits that are available analytically, and then introduces Level 0, Level 1, and Level 2 treatments of spectral knowledge. The worked examples use the same four-component dataset throughout so that changes in the results can be traced directly to changes in prior information. Particular attention is given to simplex geometry, endmember bias, mixture diversity, covariance, cross-location dependence, and uncertainty propagation.

Organization. Sections 1–2 define the materials problem and the relevant prior work. Sections 3–4 introduce the general formulation and the analytical cases that can be solved exactly or locally. Sections 5–7 use one four-component dataset to compare Levels 0–2 and to expose covariance and posterior dependence. Section 8 gives the experimental data contract, while Section 9 summarizes the extensions required for phase maps, solid solutions, spatial coupling, and counting instruments.

1. Materials Setting and Use Cases

1.1 Spectral Mixtures in Materials Characterization

Consider a probe volume at location i that contains K distinguishable material components. A component can mean a phase, an oxidation state, a coordination environment, an emitting centre, or any other state whose intrinsic spectrum is reproducibly defined. Let f_{ik} be the physical fraction of component k in that probe volume, let η_k be its spectral yield per unit physical amount, let $s_k(E)$ be its unit-area intrinsic spectral shape, and let A_i collect exposure, thickness, beam current, collection geometry, or any other location-dependent scale. The expected intrinsic signal before detector response is then

$$r_i(E) = A_i \sum_{k=1}^K f_{ik} \eta_k s_k(E) \quad (1)$$

Equation (1) is the materials starting point. It makes an important distinction that is often hidden when a matrix factorization is written directly: the fraction of the measured signal is not automatically the mass, atomic, or volume fraction of the material. If the yields η_k differ, the normalized spectral mixture weights are

$$w_{ik} = \frac{f_{ik} \eta_k}{\sum_j f_{ij} \eta_j} \quad (2)$$

and only when the relevant yields are equal, calibrated, or absorbed into the component definition does $w_{ik} = f_{ik}$. The present Stage-1 model adopts that simplification: composition and mixture fraction are identified. This is a baseline assumption, not a universal statement about spectroscopy.

After removing an overall scale or working with an intensity convention in which the endmembers are normalized consistently, the local spectrum becomes

$$y_i(E) = \sum_{k=1}^K w_{ik} s_k(E) + \epsilon_i(E) \quad (3)$$

where $\epsilon_i(E)$ is measurement noise. The fractions are constrained by

$$w_{ik} \geq 0, \quad \sum_{k=1}^K w_{ik} = 1 \quad (4)$$

1.2 Composition, Spectral, and Measurement Information

In the materials problems of interest, Eq. (3) is accompanied by two additional data streams. First, an independent assay supplies imperfect information about the fractions. In the present Stage-1 setting this can be written schematically as

$$z_i^{(C)} = w_i + \delta_i \quad (5)$$

with the important qualification that a closed composition vector should be modelled in simplex-native coordinates rather than by independent Gaussian errors on all fractions. Reference measurements, databases, physical line-shape models, simulations, or libraries supply information about the component spectra. The inference therefore combines the mixture spectra Y , composition evidence $z^{(C)}$, and spectral evidence $z^{(S)}$, each with its own uncertainty model.

DIFFERENT EXPERIMENTS CAN SUPPLY THE SAME THREE INFORMATION CHANNELS

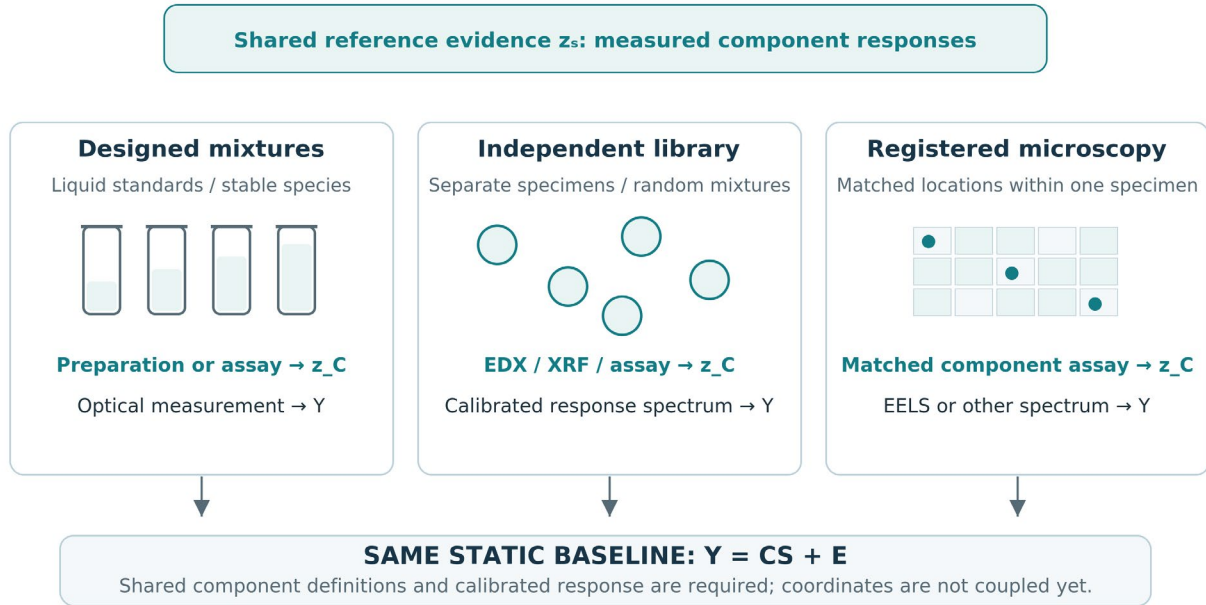


Figure 1. Three experimental architectures can supply the same basic information streams: designed mixtures, an independent library of specimens, or registered microscopy measurements at matched locations.

ONE SHARED SPECTRAL MATRIX; MANY MIXTURE COMPOSITIONS

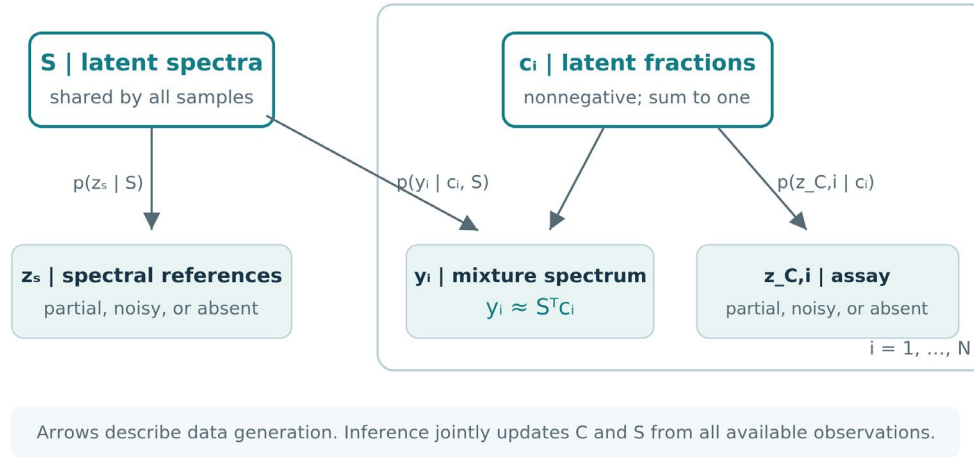


Figure 2. Stage-1 generative structure: one globally shared spectral matrix, many local mixture compositions, and separate evidence channels for composition and spectral references.

1.3 Representative Materials Use Cases

The Stage-1 model appears naturally in several materials experiments. The examples below differ in instrumentation, but the information structure is the same: a local mixture spectrum is measured together with partial knowledge of composition and partial knowledge of component spectra.

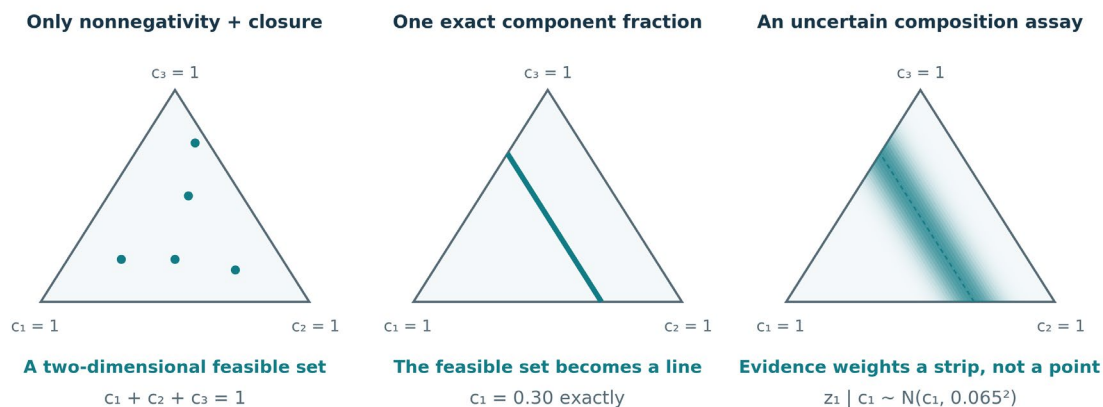
Materials setting	Where the two information streams come from	Most natural model
Combinatorial thin-film library	EDX/XRF or synthesis record gives local composition; Raman, cathodoluminescence, optical, EELS, or diffraction gives a local spectrum.	Level 0 if reference spectra are trusted; Level 1 if peak positions/widths vary; Level 2 if reference or simulation ensembles are available.
Diffusion couple / composition gradient	Composition changes continuously across position and can be measured independently while a second spectroscopy probes local bonding or structure.	The diversity of compositions is an asset: it separates global endmembers if the mixtures span independent directions.
STEM spectrum image	Each pixel can contain several phases or chemical environments; EELS/EDX provide spectra and a co-registered assay may constrain fractions.	Counting statistics, thickness, and detector response eventually need explicit modelling; Stage 1 is the Gaussian baseline.
Phase mapping by diffraction or vibrational spectroscopy	A local diffraction/Raman spectrum is a mixture of phase signatures; synthesis or elemental analysis constrains plausible fractions.	Functional priors are attractive when line shapes are not reliably captured by a few named parameters.
Scanning-probe spectroscopic imaging	Local I–V, resonance, or other spectroscopy may mix responses of unresolved regions or states.	The same mathematics applies only when linear superposition is physically reasonable; otherwise the forward model must be generalized.
Prepared mixtures / standards	Known preparation fractions and measured mixtures calibrate unknown spectra or validate uncertainty models.	This is the cleanest route to testing coverage and identifiability because composition can be independently controlled.

1.4 Distinct Sources of Uncertainty

Measurement noise is variation of a new measurement around a fixed latent mixture. Composition uncertainty is incomplete knowledge of the local fractions. Spectral uncertainty is incomplete knowledge of the global component functions. These are different random objects. A fourth distinction is equally important: epistemic spectral

uncertainty means there is one unknown component function shared by all locations; intrinsic variability means each location has a genuinely different realization. More mixture measurements can reduce epistemic uncertainty, but they do not average away intrinsic variability.

CLOSURE CHANGES THE GEOMETRY OF PARTIAL INFORMATION



Illustrative geometry. Shading is an assay likelihood on the simplex, not a fitted posterior or experimental result.

Figure 3. Composition uncertainty lives on a simplex, not in a Cartesian box of independent error bars. Fixing one fraction restricts the feasible set of the others; uncertain composition produces a band across the simplex.

1.5 Scope of the Stage-1 Model

The Stage-1 model assumes one globally shared spectrum per component, linear mixing, a known number of components, no additional baseline or gain drift, and exchangeable measurement locations. Composition is identified with mixture fraction. Spatial and temporal coupling, solid solutions, composition-to-phase maps, nonlinear mixing, unknown components, explicit instrument response, and Poisson counting are treated as extensions. This restricted model is useful because several limits are analytically tractable and can be used to test the numerical inference.

2. Prior Work and Related Approaches

The relevant prior work is distributed across several communities. Chemometrics developed soft and hard constraints on concentrations and spectra. Receptor modelling made partial guidance on both factors an operational tool. Remote sensing developed models of spectral variability. Quantitative NMR provides the closest analogue of parametric spectral priors. Gamma spectrometry combines generative spectral models with counting statistics and interval validation. Electron microscopy supplies the materials-mapping setting in which these ideas can be combined with independently measured chemistry. The point is not to rename these methods, but to identify which parts of each are needed for a given materials experiment.

2.1 Chemometrics and multivariate curve resolution (MCR)

MCR is the most developed language for a bilinear concentration–spectrum model. Gemperline and Cash formalized the distinction between hard constraints and soft penalty terms, explicitly arguing that experimentally supplied information is rarely exact [1]. Known-value constraints then fixed selected entries of either factor [2], while the soft known-value extension quantified what happens when the supposedly known values are wrong [3]. Hard–soft modelling couples mechanistic kinetic or equilibrium models to part of the concentration matrix [4]. MCR-BANDS and area-of-feasible-solutions methods go beyond a single optimum and ask which alternative factorizations remain admissible under the constraints [5,6].

For the present tutorial the important lesson is that “partial knowledge” has at least three independent attributes: coverage (which entries or directions are informed), precision (how tightly), and bias (how wrong the centre may be). Merely multiplying one covariance matrix by a scalar explores precision along a fixed set of directions; it does not explore missing or misdirected knowledge.

2.2 Receptor modelling and source apportionment

Positive matrix factorization and the multilinear engine (ME-2) provide perhaps the closest operational analogue of dual soft information [7–9]. ME-2 uses an α -value to constrain a factor profile or time series to remain within a fractional tolerance of a supplied reference, and the same mechanism can be applied on both factors. In this community, uncertainty estimation is part of routine analysis: bootstrap addresses random error, displacement probes rotational ambiguity, and combined procedures assess both [10]. Bayesian multivariate receptor models treat source profiles and contributions jointly, and later work adds temporal or spatial structure [11–13]. BAMF is especially close to the present framework because it uses a full Bayesian factorization with temporal coupling and demonstrates the value of partial prior information on component composition [14].

The transferable lesson is practical: a tunable prior width on both factors is already a standard operational object. What the materials problem adds is the need to combine that idea with simplex-aware assays, physically structured spectral priors, detector-specific likelihoods, and co-registered materials measurements.

2.3 Informed NMF

Informed NMF takes a different algorithmic route. Known entries of a factor are substituted exactly, and only the unknown entries are optimized [15,16]. This is mathematically related to an infinite-precision prior, but the reparameterized algorithm behaves better than simply adding an enormous penalty. The lesson is important for software design: exact knowledge should usually reduce the parameterization rather than be represented numerically as an extremely narrow Gaussian.

2.4 Hyperspectral remote sensing

Remote sensing has the richest literature on spectral variability. Fully constrained least squares provides the known-endmember limit [17]. Bayesian formulations and the normal compositional model treat endmember variability probabilistically [18,19]. Perturbed and extended linear mixing models introduce location-dependent spectral changes [20,21], while endmember bundles and deep generative models represent more flexible manifolds [22,23]. Spatial priors on abundances are also well developed. The asymmetry is that remote sensing usually has no independent abundance assay at the sub-pixel scale; “semi-supervised unmixing” therefore usually means spectral-library guidance, not measured fractions. That is why the dual-information problem was solved more completely in communities where mixtures can be prepared or independently assayed.

2.5 Quantitative NMR and metabolomics

Quantitative NMR is the cleanest precedent for Level 1. BATMAN and related Bayesian models infer concentrations together with uncertain shifts, widths, phases, and template parameters under priors [24–26]. This literature teaches two key ideas. First, a physical parameter such as a peak shift generates a correlated deformation across channels; independent channelwise error bars are the wrong object. Second, a physically parameterized spectrum may require only a handful of unknowns instead of one free intensity at every channel, dramatically changing identifiability.

2.6 Nuclear and gamma-ray spectrometry

Gamma spectrometry is the strongest precedent for the combination of generative spectral priors, Poisson counting, and calibrated uncertainty. SEMSUN learns a low-dimensional manifold of signature deformations and uses it in semi-blind Poisson unmixing [27]. The Bayesian uncertainty extension compares Laplace/Fisher and MCMC uncertainty and validates interval coverage against metrological expectations [28]. Lessard et al. provide a recent example of NMF with partial prior knowledge on both factors [29]. For dose-limited electron and X-ray spectroscopy, the transferable lesson is that low counts should be modelled as counts: zeros are data, background subtraction changes the likelihood, and coverage on repeated realizations is the decisive test.

2.7 Analytical electron microscopy

Electron microscopy has imported many of the algorithms but not yet the complete dual-information workflow. Bayesian spectral mixture analysis has been applied to EELS spectrum images [30]; sparse Poisson-aware NMF has been used for EELS and EDX [31]; co-registered EELS/EDX can separate heavily overlapping components [32]; and physics-guided NMF for STEM-EDXS builds a physical phase model directly into the factorization [33]. Multimodal physically constrained linear unmixing is another close precedent [34]. What remains unusual in the intended microscopy instance is the simultaneous use of an external composition assay at matched sampling volumes and uncertain reference spectra with fully propagated covariance.

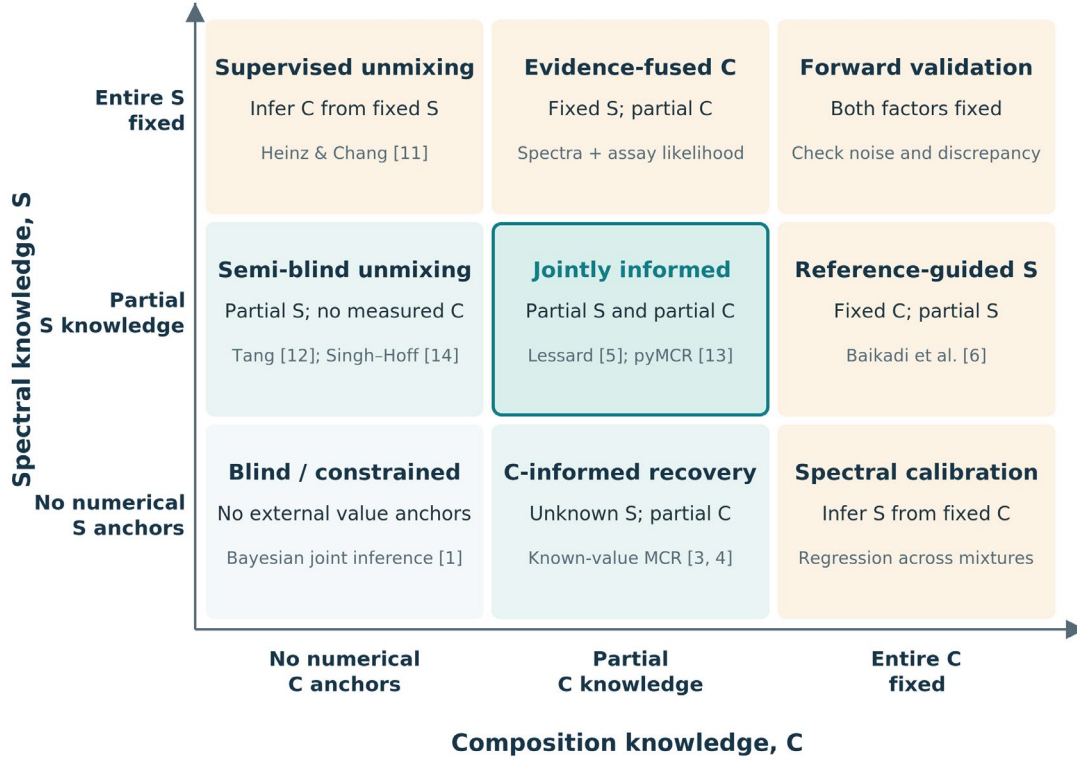
2.8 Relation Between the Different Approaches

Community	What it already contributes	Typical likelihood	What to borrow
Chemometrics / MCR	Soft/hard constraints on both factors; feasible-solution geometry	Often Gaussian least squares	Constraint strength, bias, rotational ambiguity
Receptor modelling	Dual guidance on profiles and contributions; temporal/spatial Bayes	Weighted least squares / Bayesian	Operational uncertainty workflows
Informed NMF	Exact substitution of known entries	NMF divergences	Hard knowledge as reparameterization
Remote sensing	Spectral variability, bundles, generative manifolds, spatial priors	Gaussian / robust variants	Rich spectral side; little independent abundance data
Quantitative NMR	Parametric line-shape priors	Gaussian-like spectral noise	Direct Level-1 analogue
Gamma spectrometry	Generative spectral priors + Poisson likelihood	Poisson	Coverage, counting metrology
Electron microscopy	EELS/EDX unmixing, multimodal fusion, physics-guided NMF	Gaussian or Poisson approximations	Materials use case; dual external information still thin

The literature can be organized along two axes: how much is known about composition and how much is known about the component spectra. The centre of this plane—soft or measured information on both factors—is already occupied by MCR, ME-2, Bayesian receptor models, and related methods. For materials spectroscopy the remaining task is experimental rather than terminological: define which measurement constrains each factor, carry the associated uncertainty into the model, and preserve dependencies when the composition and spectral evidence share calibration or provenance.

THE COMPOSITION-SPECTRAL KNOWLEDGE PLANE

Current setting: one shared S ; no explicit spatial or temporal coupling



Within “partial”: bounds (B) · fixed subsets (H) · soft references (R) · uncertain measurements (M)
 Qualitative constraints (Q) are an overlay. Positions are categorical, not information scores.

Figure 4. The composition–spectral knowledge plane. The important region for this tutorial is the centre: both composition and spectra are partially known. The surrounding cells are not different theories; they are limiting information regimes.

What is not the claim. It is not novel that constrained regression, NMF/MCR, and Bayesian factorization are limits of one bilinear model with different prior widths. That continuity is useful as a tutorial and as a software test. The more substantive questions are whether supplied uncertainty is admissible, whether the evidence streams are independent, whether intervals are calibrated, and whether the model matches the microscopy or materials experiment.

3. General Formulation

3.1 Linear Mixing Model

Stack N measured spectra as rows of $Y \in \mathbb{R}^{N \times M}$, local fractions as rows of $W \in \mathbb{R}^{N \times K}$, and component spectra as rows of $S \in \mathbb{R}^{K \times M}$. Equation (3) becomes

$$Y = WS + E \quad (6)$$

where E is measurement noise. The row constraints in Eq. (4) make every W_i a point on the K -component simplex. Composition evidence enters through a likelihood or prior for W , and spectral knowledge enters through a model or prior for S . For Gaussian measurement noise with covariance Σ_y and prior blocks $p(W)$ and $p(S)$, the posterior has the schematic form

$$p(W, S | D) \propto L_Y L_C L_S \pi_W \pi_S \quad (7)$$

Here $D = \{Y, z^{(c)}, z^{(s)}\}$. Equation (7) makes the modular structure explicit. It also exposes an assumption that must be checked experimentally: the composition evidence $z^{(c)}$ and spectral evidence $z^{(s)}$ are conditionally independent once W and S are known. If they share calibration data or one was computed using the other, the factorization overcounts information unless the shared nuisance is modelled.

3.2 Composition on the Simplex

A convenient way to enforce Eq. (4) is to work in $K - 1$ additive-log-ratio coordinates. Taking component K as the reference,

$$z_{ij} = \log\left(\frac{w_{ij}}{w_{iK}}\right), \quad w_i = \text{softmax}(z_i, 0) \quad (8)$$

Gaussian uncertainty in z is unconstrained and maps to a valid distribution on the simplex. It is not equivalent to assigning independent Gaussian errors to the raw fractions. Near zero fractions the transformed distribution becomes skewed, so effective uncertainty in physical fraction space should be recomputed from draws rather than inferred from a single linear error bar.

3.3 Three Levels of Spectral Knowledge

Three levels of spectral knowledge used in this work

Level 0 — exact endmembers

The component spectra $S_k(E)$ are fixed. Spectral uncertainty is zero by assumption. Use this level only when reference spectra can be treated as exact on the scale relevant to the inference.

Level 1 — parametric spectral uncertainty

Each spectrum is written as $S_k(E) = g_k(E; \theta_k)$, with priors on physically named parameters such as peak areas, positions, widths, onsets, line ratios, or phases.

Level 2 — functional spectral uncertainty

Each endmember is an uncertain complete function represented in a reduced basis learned from replicates, simulations, or a spectral library. Correlations across spectral coordinate are retained.

The level specifies how spectral knowledge is represented; it does not rank the amount of information.

Figure 5. The three spectral-knowledge levels used in this work. Level 0 treats the endmember spectra as exact; Level 1 represents spectral uncertainty through physically named parameters; Level 2 represents uncertainty in complete spectral functions and retains covariance across spectral coordinate.

The complete Stage-1 map: three concentration-prior regimes by three spectral-prior regimes

Uncertainty of mixing weights, τ_w	Narrow	Forward validation $W = C, S = S_0$	Bayesian spectral calibration $W = C, \text{infer } S$	Ordinary spectral calibration $S = C^+ Y$
	Finite	Bayesian concentration regression $S = S_0, \text{regularize } W$	Informed semi-blind inference $\text{infer } W \text{ and } S$	Composition-anchored semi-blind $\text{prior on } W \text{ only}$
	Broad	Constrained regression / FCLS $W = Y S_0^T (S_0 S_0^T)^{-1}$	Spectrum-anchored semi-blind $\text{prior on } S \text{ only}$	Blind factorization NMF / MCR / BSS
		Narrow	Finite	Broad
		Uncertainty of endmember spectra, τ_s		

Figure 6. The Stage-1 regime map. Narrow or broad composition and spectral priors move the same bilinear model between forward prediction, regression/calibration, informed semi-blind inference, and blind factorization.

Level 0 fixes S exactly. Level 1 writes each spectrum as $S_k(E) = g(E; \theta_k)$ with a prior on physically named parameters such as peak areas, positions, widths, onsets, or phases. Level 2 represents each endmember as a random function, usually in a reduced basis learned from replicates, simulations, or a library. These levels describe the form of spectral knowledge, not its numerical strength: a narrow Level-1 prior can be more informative than a broad Level-2 prior, while a Level-2 basis that omits a real deformation direction can be more restrictive than any prior-width choice.

4. Analytical Cases

4.1 Level 0: known endmembers

For one location, write $y = S^T w + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \Sigma_y)$. Suppose the external composition assay provides a Gaussian prior $w \sim \mathcal{N}(c, \Sigma_c)$ before enforcing the simplex boundary. Completing the square gives the exact unconstrained Gaussian posterior covariance

$$\Sigma_{w|y} = (S \Sigma_y^{-1} S^T + \Sigma_c^{-1})^{-1} \quad (9)$$

and posterior mean

$$\mu_{w|y} = \Sigma_{w|y} (S \Sigma_y^{-1} y + \Sigma_c^{-1} c) \quad (10)$$

Equations (9)–(10) show the information balance directly. The spectroscopy contributes $S \Sigma_y^{-1} S^T$; the external assay contributes Σ_c^{-1} . Exact closure is handled by transforming to $K - 1$ independent coordinates or by conditioning the Gaussian on $\mathbf{1}^T w = 1$. Active non-negativity then truncates the Gaussian and can shift its mean near a boundary.

The eigenvalues of $S \Sigma_y^{-1} S^T$ in the simplex tangent space quantify spectral separability. If two endmembers become similar, one or more eigenvalues approach zero and the corresponding fraction contrast is not identified by the spectra.

4.2 Bias from Imperfect Fixed Endmembers

Level 0 is reliable only when the fixed endmembers are correct on the scale required for the composition estimate. In the broad composition-prior limit, weighted least squares with an assumed spectral matrix $\hat{S}$ gives

$$\hat{w} = (\hat{S}\Sigma_y^{-1}\hat{S}^T)^{-1}\hat{S}\Sigma_y^{-1}y \quad (11)$$

With $P_{\hat{S}}$ denoting the weighted pseudoinverse in Eq. (11), if the data are generated by S_* rather than $\hat{S}$, the mean bias is

$$\mathbb{E}[\hat{w} - w_*] = P_{\hat{S}}(S_*^T - \hat{S}^T)w_* \quad (12)$$

Equation (12) explains why an apparently small spectral mismatch can produce a systematic composition error. Only the part of the spectral error that projects through the weighted pseudoinverse matters, so a reconstruction can look excellent while the inferred fractions are biased. This is the formal motivation for Levels 1 and 2.

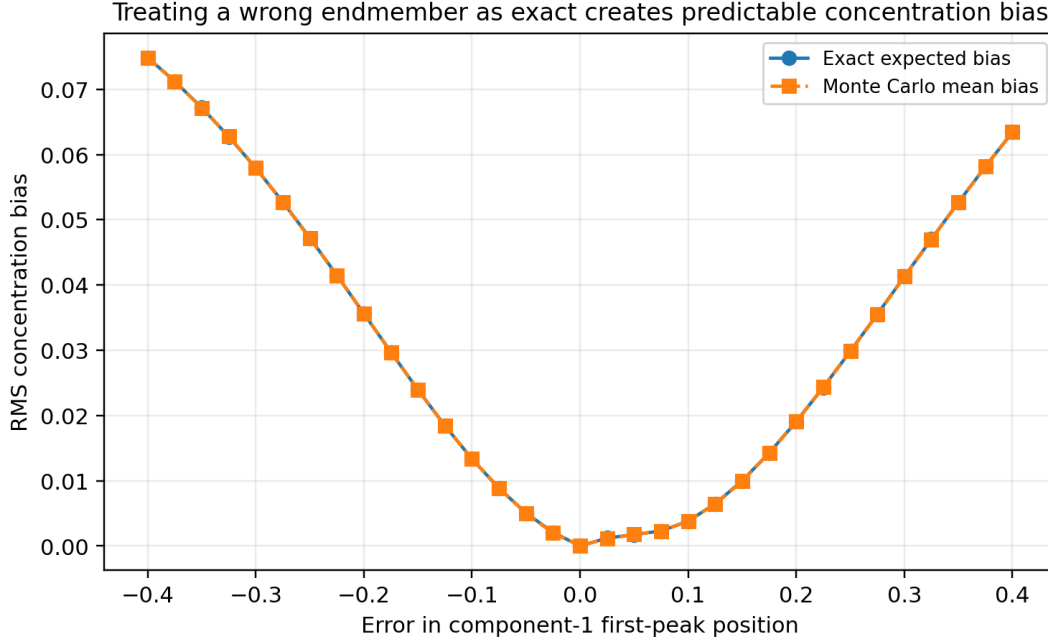


Figure 7. Exact and Monte Carlo composition bias produced by shifting one endmember peak while treating the shifted spectrum as exact. A visually modest reference error can create a systematic fraction bias.

4.3 Known Compositions and Uncertain Spectra

If W is known and the spectra are uncertain, the problem becomes ordinary multivariate calibration. At channel m , let y_m be the N -vector of measurements across mixtures and s_m the K -vector of component intensities. With prior $s_m \sim \mathcal{N}(s_{0m}, K_{0m})$ and Gaussian noise $\sigma_m^2 I$,

$$K_{m|y} = (K_{0m}^{-1} + \sigma_m^{-2} W^T W)^{-1} \quad (13)$$

so the information grows through $W^T W$. Repeating one composition many times strengthens only one weighted combination of spectra; compositionally diverse mixtures increase rank and separate additional endmember directions. Thus mixture diversity can be more important than repeated measurements at one composition.

MORE MEASUREMENTS ARE NOT THE SAME AS MORE COMPOSITION DIVERSITY



Constructed three-component examples; no pure samples. Rank is necessary for unrestricted known-C regression.

Figure 8. More measurements are not equivalent to more composition diversity. Repeated mixtures can leave the design rank deficient, whereas independent composition contrasts separate the endmembers.

4.4 Level 1: parametric spectral uncertainty

Let $S_k(E) = g(E; \theta_k)$. Around a nominal parameter vector θ_0 ,

$$S(E; \theta) \approx S(E; \theta_0) + J_\theta(E) \delta\theta \quad (14)$$

With parameter covariance C_θ , the induced spectral covariance is

$$K_S(E, E') \approx J_\theta(E) C_\theta J_\theta(E')^T \quad (15)$$

Equation (15) is much more informative than pointwise error bars because the sign structure identifies the physical deformation. For a peak-position uncertainty, intensities on opposite flanks are anticorrelated; for a width uncertainty, centre and tails oppose one another; amplitude uncertainty stays one-signed across the peak. This is why the NMR literature is such a direct precedent for materials spectra with uncertain peak positions and widths.

Exact energy-energy correlation induced by peak-position uncertainty

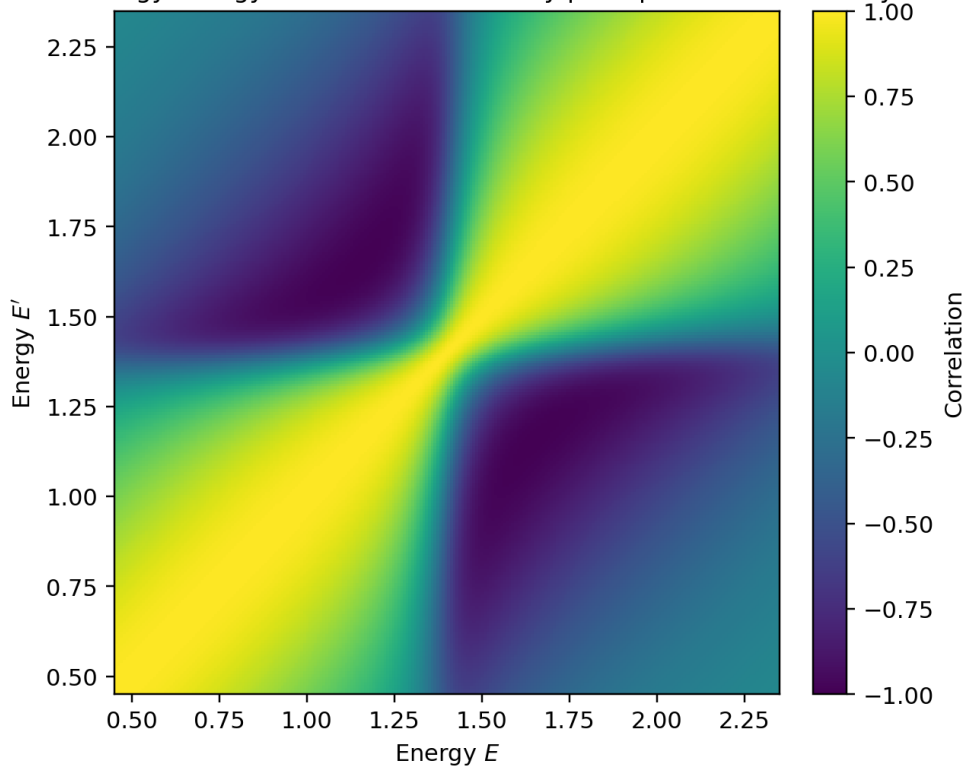


Figure 9. Exact energy–energy correlation produced by uncertain Gaussian peak position. The two flanks are anticorrelated; a pointwise uncertainty band discards this diagnostic sign structure.

4.5 Level 2: functional spectral uncertainty

When a named line-shape model is too restrictive, let $a_k \sim \mathcal{N}(0, I)$ and represent the latent log spectrum by a low-dimensional basis

$$u_k(E) = m_k(E) + B_k(E)a_k \quad (16)$$

and transform it to a positive, unit-area physical spectrum

$$S_k(E) = \frac{\exp[u_k(E)]}{\int \exp[u_k(E')] dE'} \quad (17)$$

The basis B_k can come from replicate pure-component spectra, simulations, or a curated library. In the implemented example it is learned by PCA in log-spectrum space. The empirical library, the Gaussian coefficient distribution of Eq. (16), and the induced nonlinear distribution of Eq. (17) are three distinct objects, and their means and covariances need not coincide.

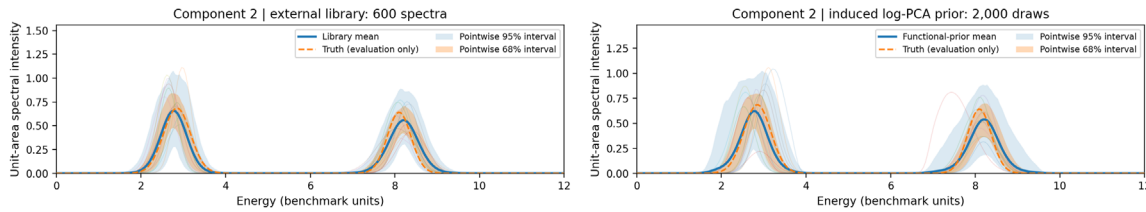


Figure 10. An empirical Level-2 library and draws from its induced log-PCA prior. Compression and Gaussianization change the distribution even though the library defines the basis.

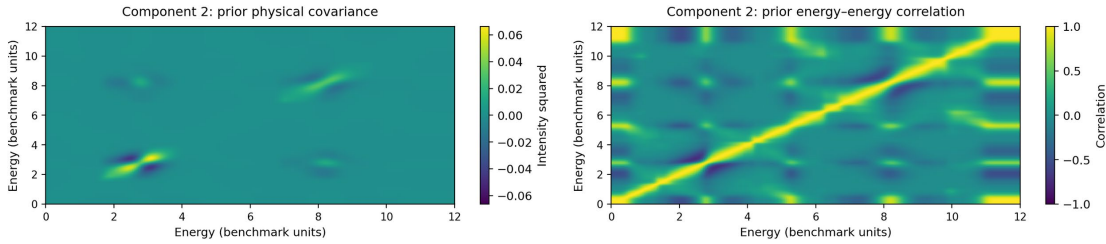


Figure 11. Physical covariance and correlation of Level-2 spectral draws. Off-diagonal structure encodes coordinated deformations; normalization forces compensating changes between spectral regions.

4.6 Shared Endmembers and Cross-Location Covariance

The Stage-1 endmember is epistemic and global: one unknown spectrum is shared by every location. If the weights are fixed and the component spectral covariances are K_k , then locations i and j have predictive cross-covariance

$$\text{Cov}(y_i, y_j | W) = \sum_{k=1}^K w_{ik} w_{jk} K_k \quad (18)$$

Equation (18) is not a spatial prior. Locations with similar compositions are correlated because they depend on the same uncertain endmember functions, even if they are far apart and measurement noise is independent. Confusing this with spatial correlation would double-count the dependence.

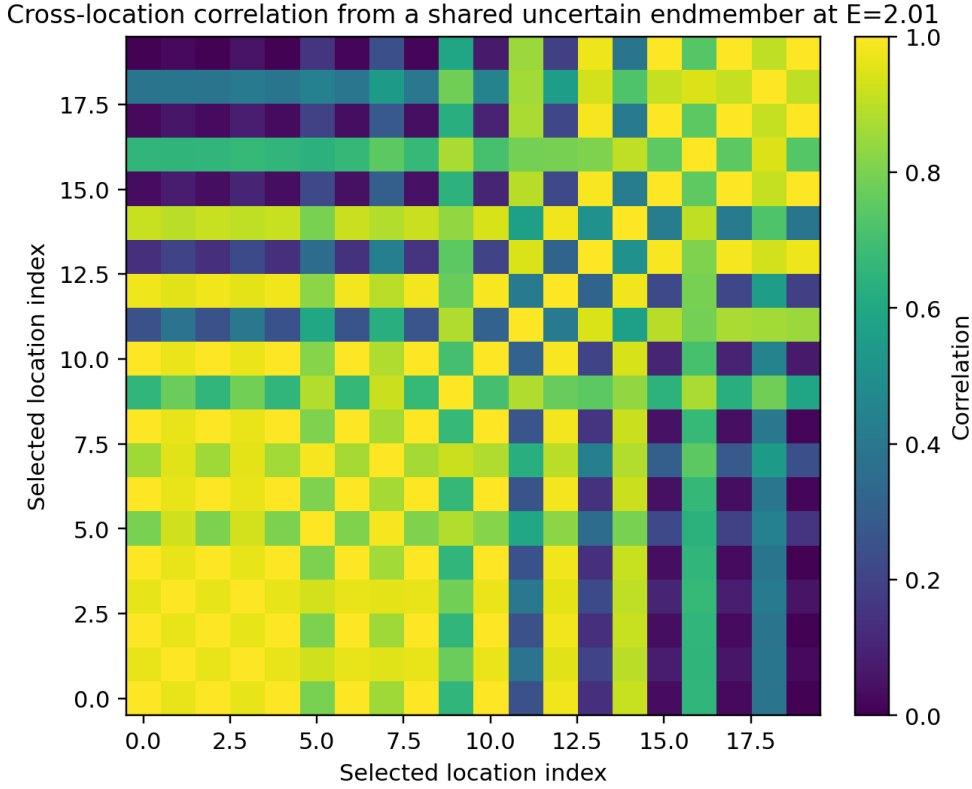


Figure 12. Cross-location correlation induced solely by shared uncertain endmembers at one spectral channel. Similar compositions inherit similar spectral uncertainty even without a spatial model.

4.7 Joint Composition–Spectral Uncertainty

When both W and S are uncertain, linearize the residual around a joint mode and partition the precision matrix into concentration and spectral blocks,

$$\Lambda = \begin{bmatrix} \Lambda_{WW} & \Lambda_{WS} \\ \Lambda_{SW} & \Lambda_{SS} \end{bmatrix} \quad (19)$$

Marginalizing spectral uncertainty gives the local concentration precision

$$\Lambda_{W,\text{marg}} = \Lambda_{WW} - \Lambda_{WS} \Lambda_{SS}^{-1} \Lambda_{SW} \quad (20)$$

The second term in Eq. (20) is positive semidefinite and subtracts information: uncertain endmembers inflate composition uncertainty relative to a calculation that fixes S at a point estimate. This is why a sequential procedure that first estimates spectra and then treats them as exact can be overconfident.

4.8 Predictive Uncertainty Decomposition

For one location write $w = w_0 + \Delta w$ and $S = S_0 + \Delta S$, where w_0 and S_0 are the posterior means. The corresponding fluctuation of the latent mixture is

$$\Delta y = \Delta w S_0 + w_0 \Delta S + \Delta w \Delta S \quad (21)$$

After inference the three terms in Eq. (21) are generally correlated, so the total variance contains their individual variances plus signed cross-covariances. Paired joint posterior draws are therefore the safest way to propagate uncertainty.

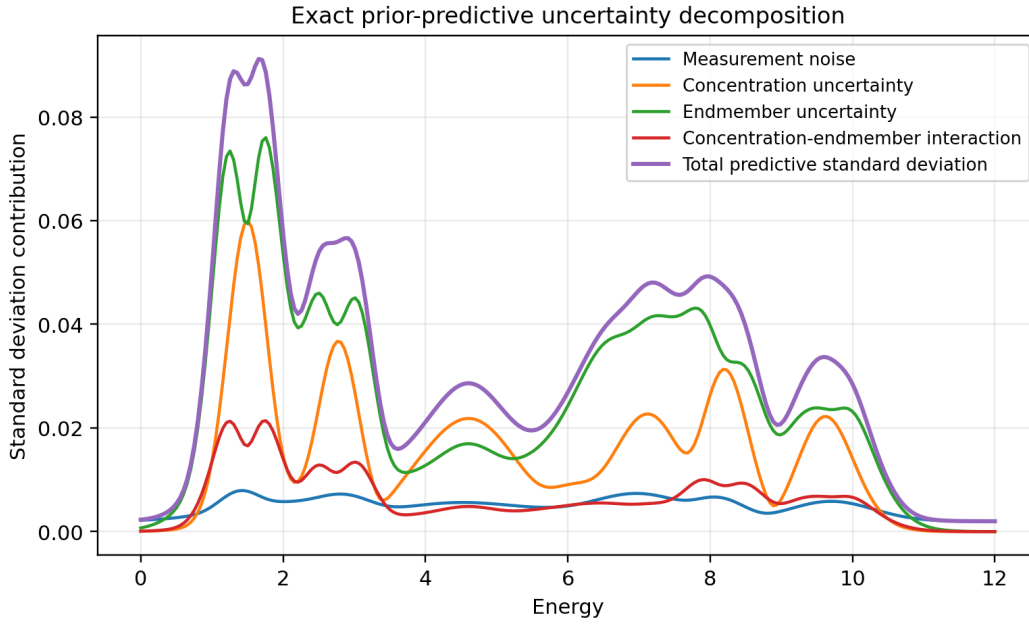


Figure 13. Energy-dependent prior-predictive uncertainty decomposition into measurement, composition, endmember, and interaction contributions. After inference the analogous decomposition contains signed dependence terms.

5. Four-Component Example

5.1 Model and Synthetic Data

The numerical tutorial uses four globally shared endmembers, each built from two peaks and normalized to unit area. The local fractions vary across measurement locations, the measured spectra are noisy linear mixtures, and the external composition information is deliberately imperfect. The same physical dataset is then analysed at Levels 0, 1, and 2. This matched construction is important: differences between levels reflect spectral knowledge, not different observations.

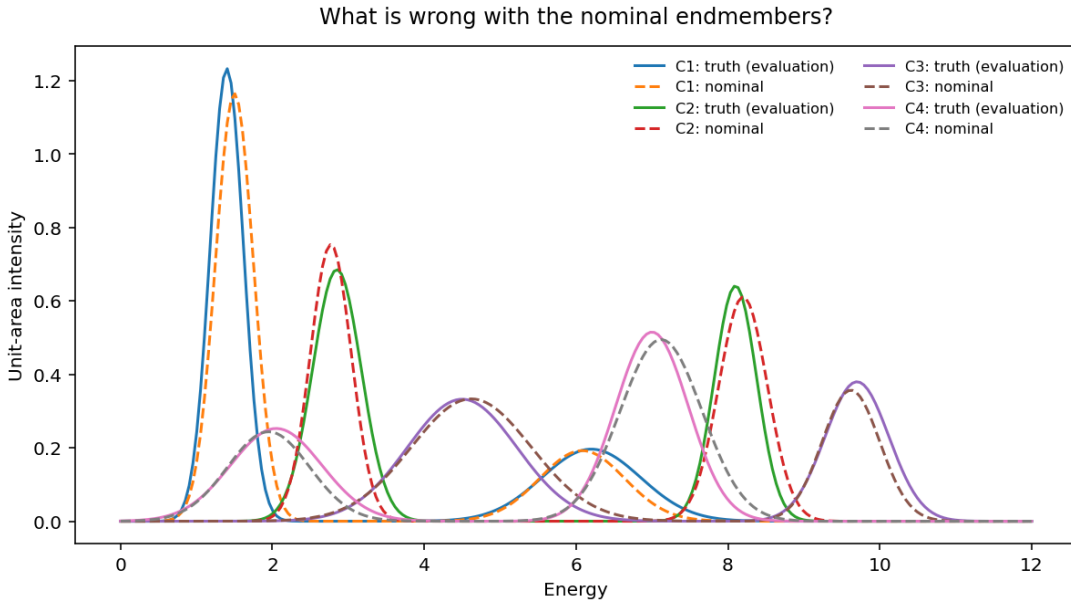


Figure 14. True endmember spectra and imperfect nominal reference spectra in the canonical four-component construction. The nominal shapes differ in area allocation, position, and width.

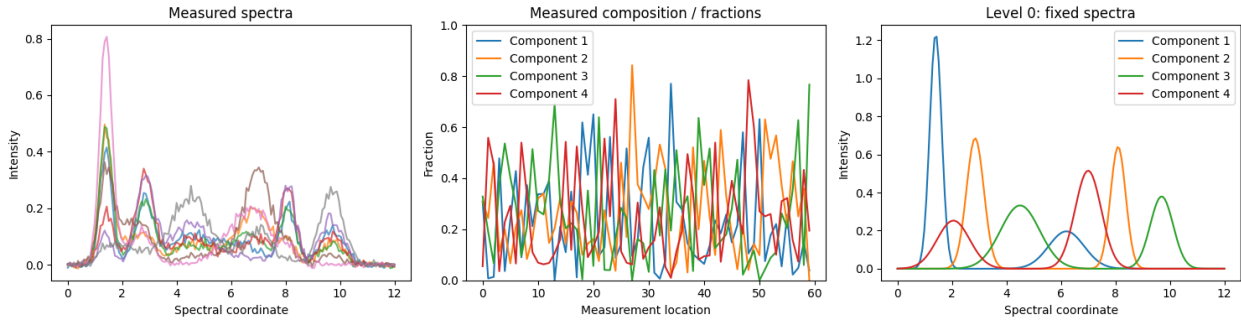


Figure 15. Executed notebook view of the common data object: representative measured spectra, measured fractions, fixed spectral references, concentration uncertainty, Level-1 parameter uncertainty, and Level-2 library uncertainty.

5.2 Level 0: Exact Endmembers

At Level 0 the only posterior uncertainty is in W . This serves as the reference benchmark: endmember error is zero by assumption, so the remaining concentration uncertainty is controlled only by the composition prior, measurement noise, and spectral separability.

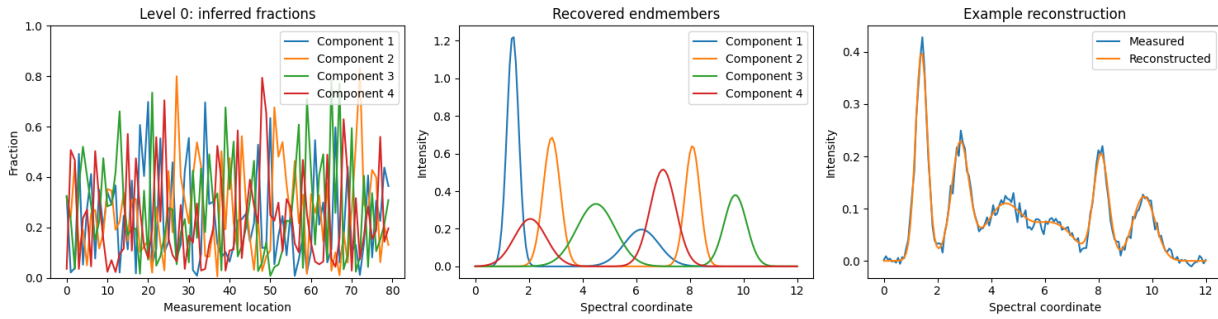


Figure 16. Level-0 notebook result: inferred fractions, exact endmembers, reconstruction, and residuals. This is the reference case against which spectral uncertainty should be judged.

5.3 Level 1: Parametric Spectral Uncertainty

At Level 1, positions, widths, and relative areas become global unknowns. The crucial test is not merely whether the reconstructed spectra improve, but whether the data can learn the biased parameter families strongly enough to avoid the Level-0 bias of Eq. (12). The canonical source study shows that allowing all relevant peak families to move is essential; freezing a biased width or position family can leave composition error an order of magnitude larger even when other parameters remain free.

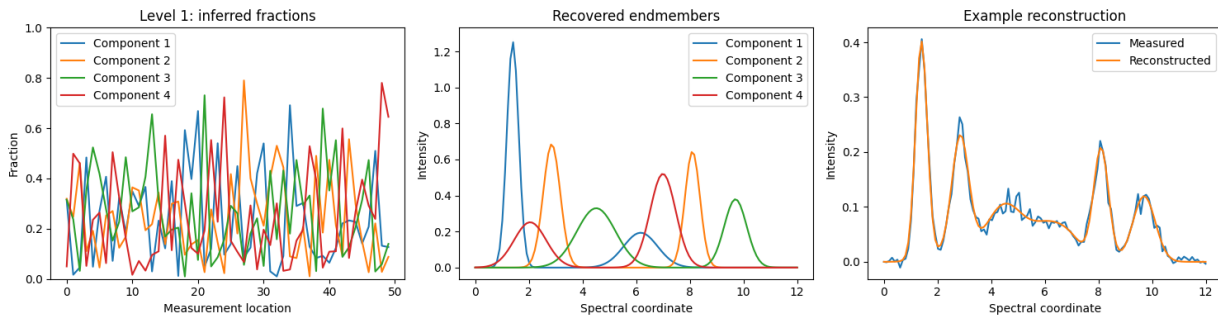


Figure 17. Level-1 posterior output from the executed tutorial.

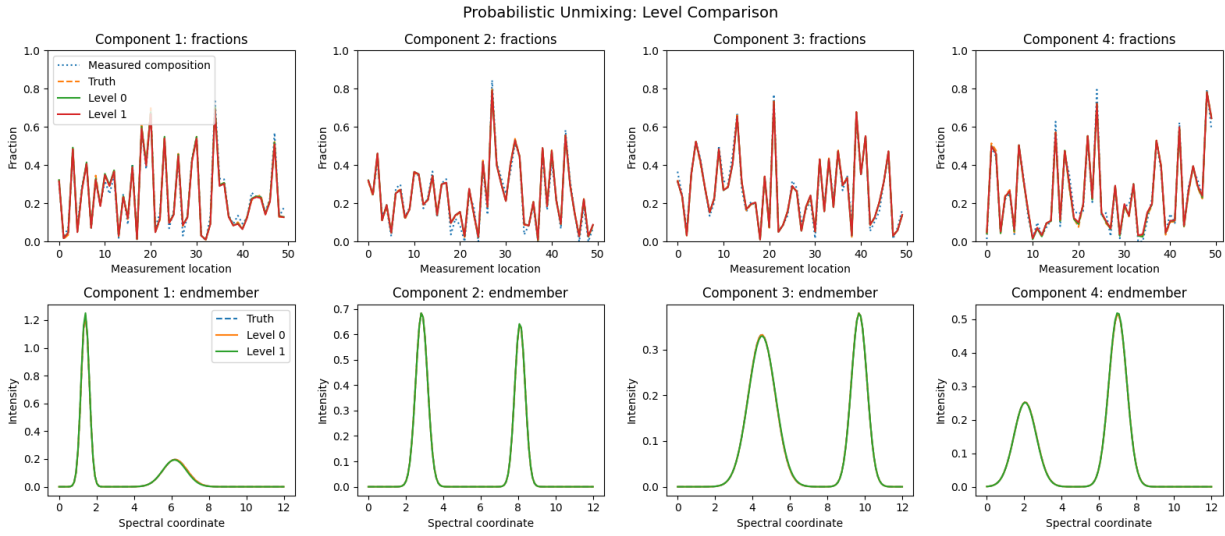


Figure 18. Direct Level-0 versus Level-1 comparison on the same physical data. The difference reflects uncertain parametric endmembers, not a different dataset.

5.4 Level 2: Functional Spectral Uncertainty

Level 2 is attractive when the analyst has an ensemble of plausible complete spectra but does not trust a small parametric family. The ensemble may come from replicates, simulation over uncertain physical inputs, or a curated library. The mixture data then update only the functional directions contained in the retained basis. Rank therefore controls the admissible function family; increasing coefficient variance cannot recover a direction that was discarded.

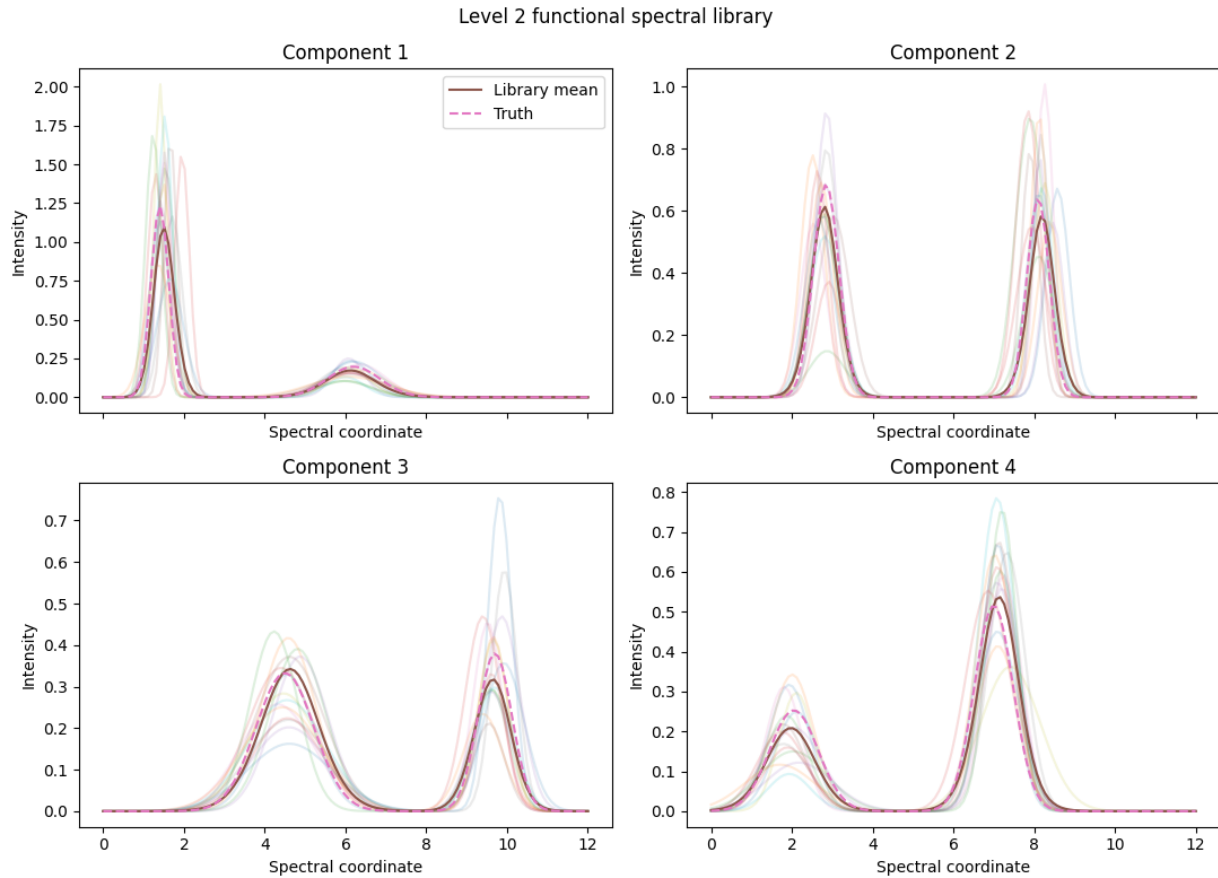


Figure 19. Level-2 notebook view of the complete spectral library for the four components.

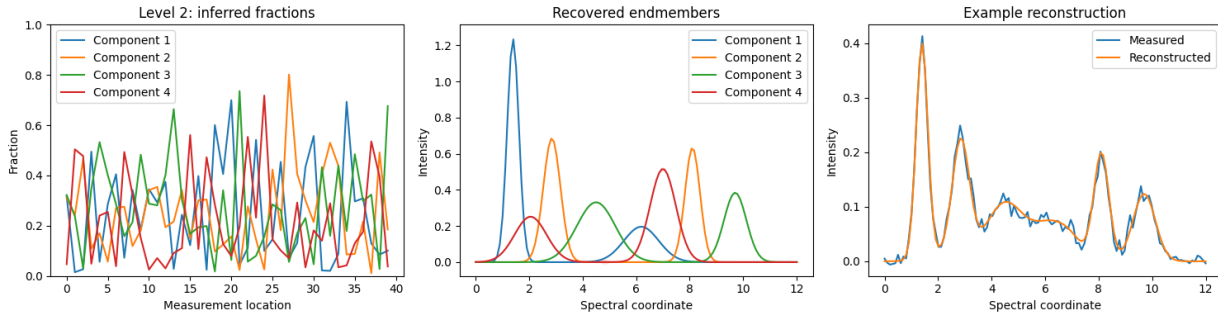


Figure 20. Level-2 posterior result: concentrations and globally shared functional endmembers are inferred jointly.

The controlled Level-2 study also shows why rank selection cannot be based on explained variance alone. Retaining 99.5% of the log-space variance did not guarantee small reconstruction error in physical intensity space. Component 2 required more modes than the selected cap and had substantially larger held-out reconstruction error than the other components. Rank therefore defines the allowed function family and must be validated in physical space.

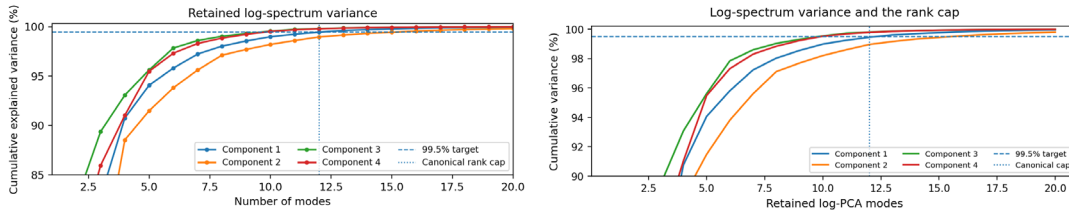


Figure 21. Cumulative retained log-spectrum variance and held-out physical reconstruction error. High variance retention can coexist with appreciable physical error, especially for difficult components.

6. Different Prior Quality for Different Components

Real materials experiments rarely provide the same quality of information for every component. One phase may have a well characterized reference spectrum but a poor composition assay, while another may have an accurately measured fraction but an uncertain spectrum. The benchmark in this section assigns uncertainty component by component so that these cases can be separated explicitly.

The executed notebook constructs four regimes: component 1 has good composition and good spectral information; component 2 has good composition and poor spectral information; component 3 has poor composition and good spectral information; component 4 is poor on both. This is the smallest experiment that reveals whether the two information streams can compensate one another.

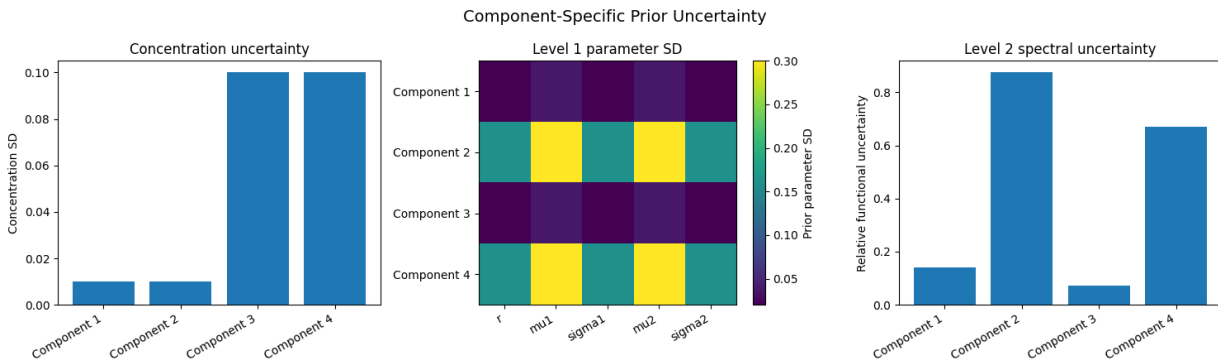


Figure 22. Input uncertainty structure for the heterogeneous benchmark. Concentration and spectral uncertainty vary independently by component.

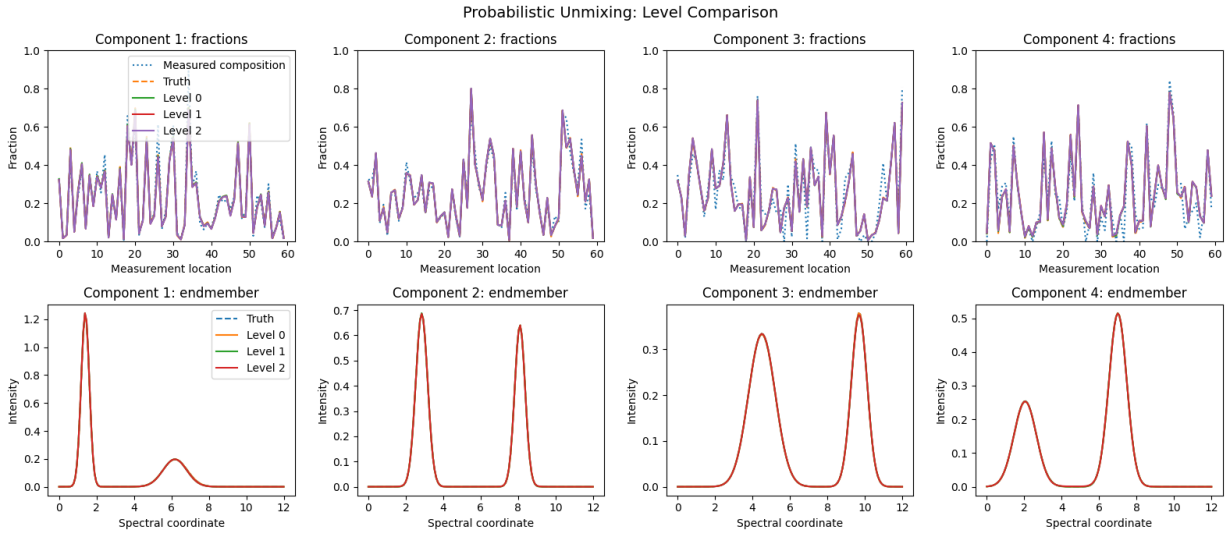


Figure 23. Level-0/1/2 comparison on the same heterogeneous dataset. Component-resolved errors and posterior widths reveal where information is coming from.

The same construction can be used to sweep the concentration uncertainty and spectral uncertainty of one component while keeping the physical truth and measured spectra fixed. The resulting maps show which information source controls the inference. Predominantly vertical variation indicates sensitivity to composition uncertainty; horizontal variation indicates sensitivity to spectral uncertainty; diagonal structure indicates a trade-off between them.

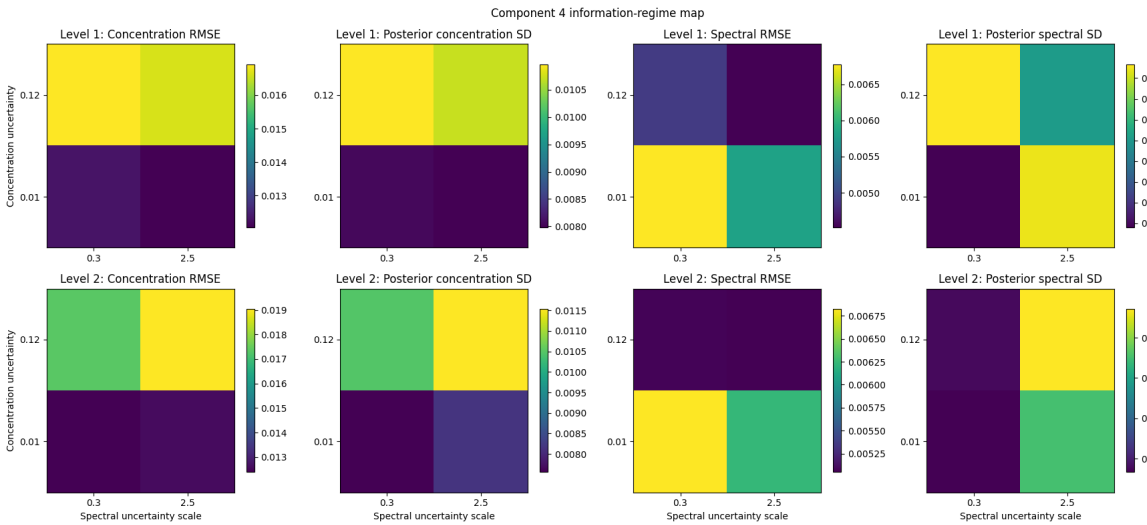


Figure 24. Executed uncertainty trade-off sweep for one component. The two axes vary prior concentration uncertainty and spectral uncertainty while the underlying physical experiment is unchanged.

7. What Spectral Covariance and Posterior Coupling Mean

7.1 Pointwise Bands, Covariance, and Correlation

A pointwise 95% band answers a marginal question at each energy. It does not say which complete functions are plausible, and it discards the sign of coordinated changes. Covariance retains absolute coupling; correlation removes the marginal scale and reveals pattern. For a shifting peak, opposite flanks are anticorrelated. For normalized spectra, gaining area in one region requires compensation elsewhere.

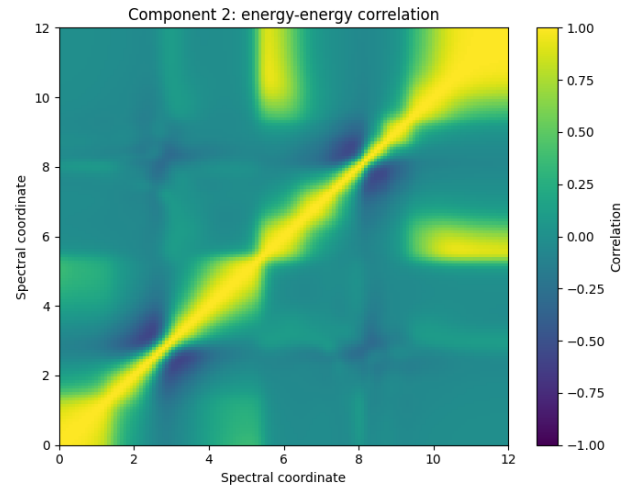
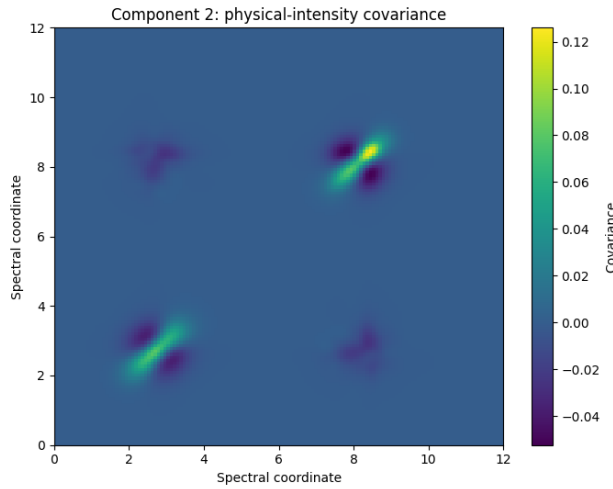


Figure 25. Physical-intensity covariance of a Level-2 component. Figure 26. The corresponding energy–energy correlation. Strong-Localized positive and negative lobes encode coordinated spectral looking correlation in weak tails should always be interpreted together with absolute SD.

7.2 Functional Modes and Physical Spectra

The latent PCA modes of Eq. (16) are signed directions in log-function space. They need not be nonnegative or normalized. A physical spectrum appears only after adding a coefficient-weighted combination of modes to the mean log function and applying Eq. (17). This distinction matters because the effect of a mode is nonlinear after normalization.

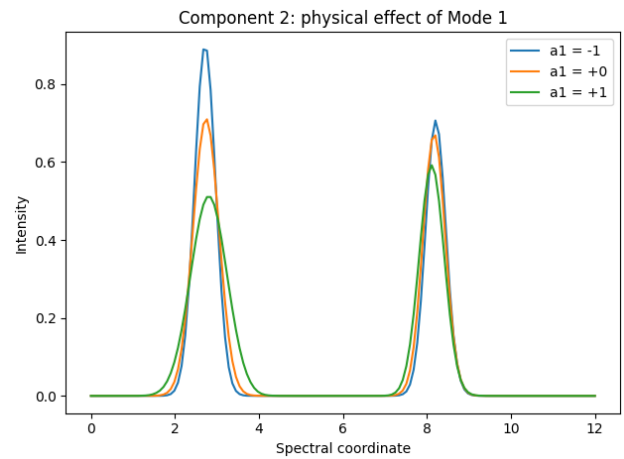
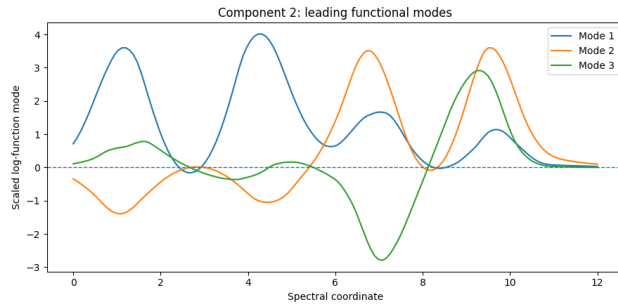


Figure 27. Leading scaled log-functional modes for one Figure 28. Physical spectra generated by changing one latent component. coefficient while holding the other coefficients fixed.

7.3 Posterior Dependence Across Components and Concentrations

Independent prior coefficients do not remain independent after fitting mixtures. Two endmembers can compensate one another, and a spectral deformation can be offset by a concentration change. The posterior covariance between blocks is therefore part of the answer. Destroying it by sampling each marginal independently gives incorrect predictive uncertainty.

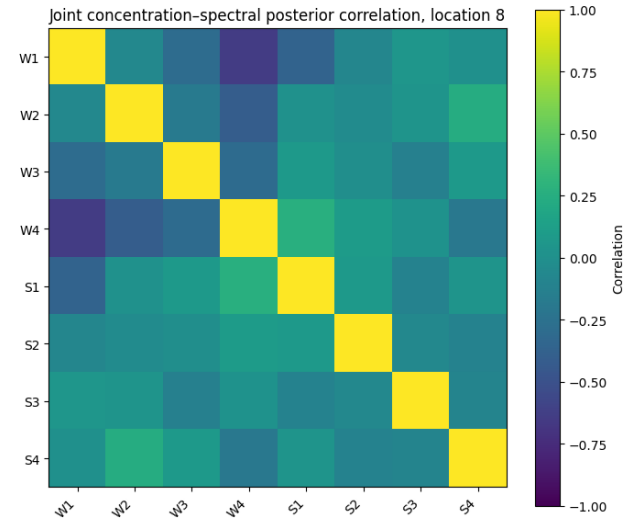
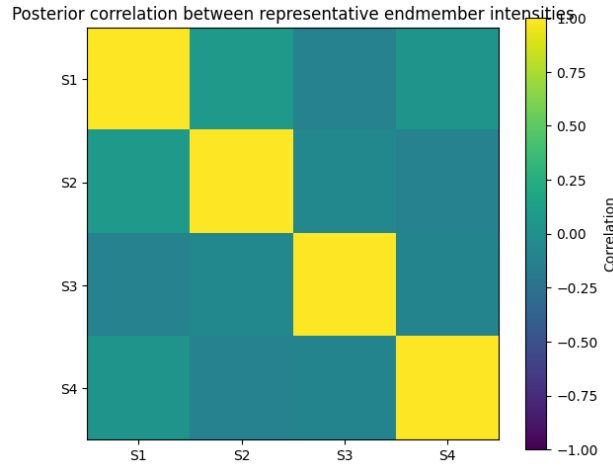


Figure 29. Posterior correlation between representative intensities Figure 30. Joint concentration–spectral posterior correlation at one of different endmembers. Off-diagonal structure is induced by mixed location. Fraction and spectral variables become coupled even though their priors were separate.

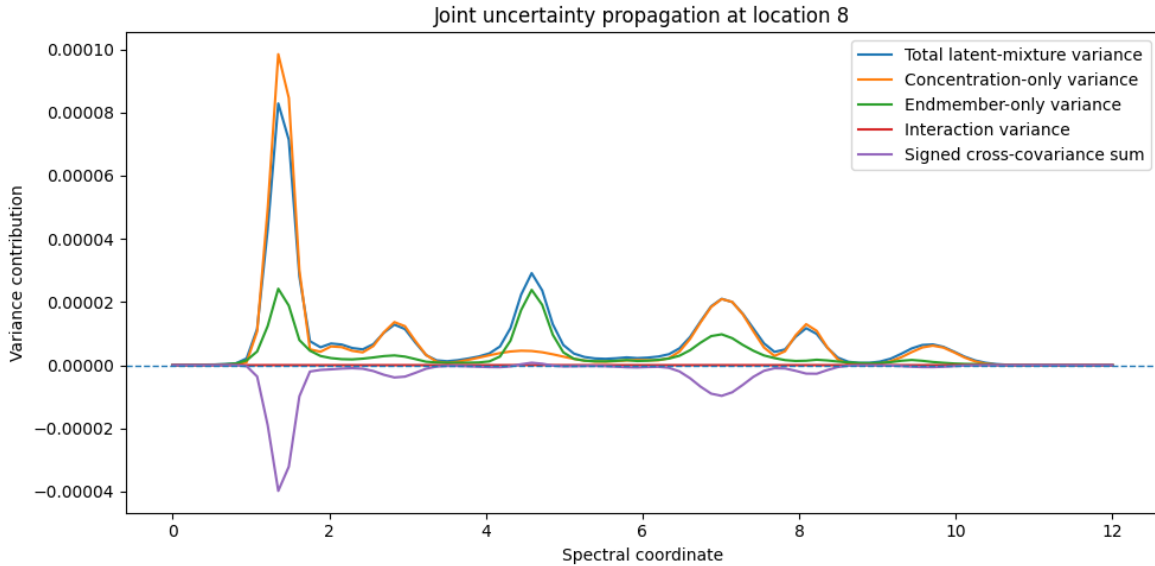


Figure 31. Exact decomposition of posterior latent-mixture variance into concentration, spectral, interaction, and signed cross-covariance terms using paired joint draws.

8. Using the Framework with Experimental Data

8.1 Required Experimental Inputs

Synthetic and experimental measurements are converted into the same data object. For the current Stage-1 case the minimum inputs are the spectral coordinate E , measured spectra Y , measured fractions W_{mean} , fraction uncertainty W_{σ} , and spectral measurement uncertainty σ_y . Spectral knowledge is then attached according to Level 0, 1, or 2.

Object	Shape	Meaning
E	$(M,)$	Spectral coordinate or channel axis
Y	(N, M)	Measured mixture spectra

W_{mean}	(N, K)	External composition/fraction estimates
W_{σ}	(N, K)	Uncertainty associated with those estimates
σ_y	scalar or (N, M)	Measurement-noise standard deviation
S_{fixed}	(K, M) , optional	Exact reference spectra for Level 0
$\theta_{\text{mean}}, \theta_{\sigma}$	(K, P) , optional	Parametric spectral prior for Level 1
S_{library}	(K, R, M) , optional	Ensemble of complete spectra for Level 2

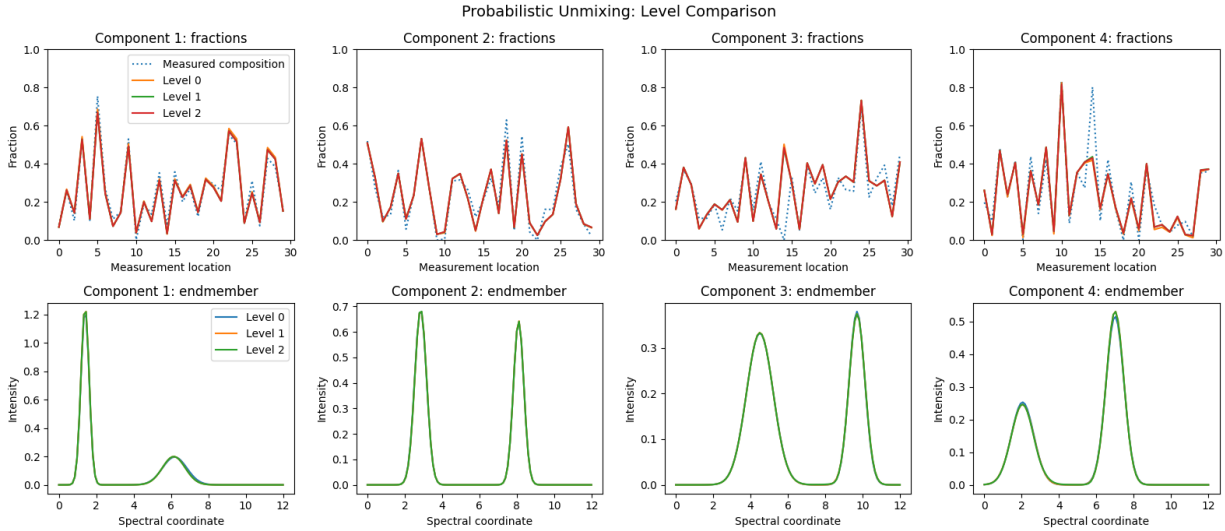


Figure 32. Experimental-style data loaded into the common data object with no synthetic truth attached.

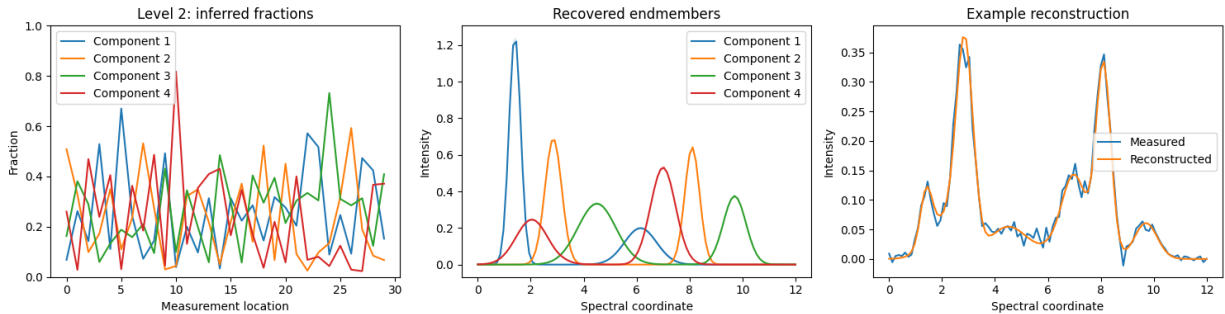


Figure 33. The same Level-0/1/2 analysis interface applied after the experimental-style data have been loaded.

8.2 Experimental Checks Before Inference

- Are the composition values actual physical fractions, or only spectral-signal fractions? If yields differ, Eq. (2) must be included.
- Do the composition uncertainties respect closure? If not, keep the raw assay model or transform to ALR/ILR coordinates rather than renormalizing error bars.
- Does the spectral library represent epistemic uncertainty in one shared endmember, or intrinsic sample-to-sample variability? They imply different models.
- Are spectra, references, and simulations on compatible energy grids and resolution conventions? Misregistration changes correlation structure, not just the mean.
- Is the noise model appropriate to the detector and preprocessing? Background subtraction, alignment, smoothing, and normalization can create channel covariance.
- Does the prior predictive distribution generate spectra resembling the observed data before fitting? If not, posterior precision is meaningless.

8.3 Choosing Level 0, 1, or 2

Level 0. Use only when the component spectra can genuinely be treated as exact on the scale of the desired fraction precision. It is excellent as an oracle or calibration benchmark and dangerous when reference mismatch is ignored.

Level 1. Use when the meaningful spectral uncertainty can be described by a small set of physical parameters—peak positions, widths, areas, phases, onsets, broadening parameters, or instrument-calibration variables.

Level 2. Use when prior knowledge is naturally an ensemble of complete spectra and the important deformation directions are better learned from replicates, simulations, or libraries than named explicitly.

9. Extensions Beyond Stage 1

9.1 Composition-to-Phase Mapping

The present tutorial sets composition equal to mixture fraction so that the role of uncertain information can be isolated. In many real materials problems the assay measures elemental fractions while the spectral components are phases, oxidation states, or coordination environments. Then the assay and spectral fractions are linked by a stoichiometric or yield map rather than identity. If two phases share the same stoichiometry, composition constrains only their sum and the spectra must separate them. If the stoichiometry is itself uncertain, it becomes another latent block.

9.2 Solid solutions and continuous endmembers

A solid solution violates the fixed-endmember assumption because the spectrum of one phase changes continuously with composition. At Level 1, line-shape parameters can be functions of composition—for example, Vegard-like shifts or disorder broadening. At Level 2, the library can be generated over composition so that the retained functional coordinate becomes a physical state variable. Distinguishing a homogeneous intermediate composition from a fine-scale two-phase mixture then becomes a model-selection problem, not merely a linear unmixing problem.

9.3 Spatial and temporal coupling

Spatial regularization is not one operation. One can couple fractions, local spectral perturbations, class labels, or residuals, and each choice represents a different physical assumption. The shared-endmember covariance of Eq. (18) already produces cross-location dependence without any spatial prior and must not be counted twice.

EXTENDING THE STATIC MODEL: KEEP KNOWLEDGE AND COUPLING SEPARATE

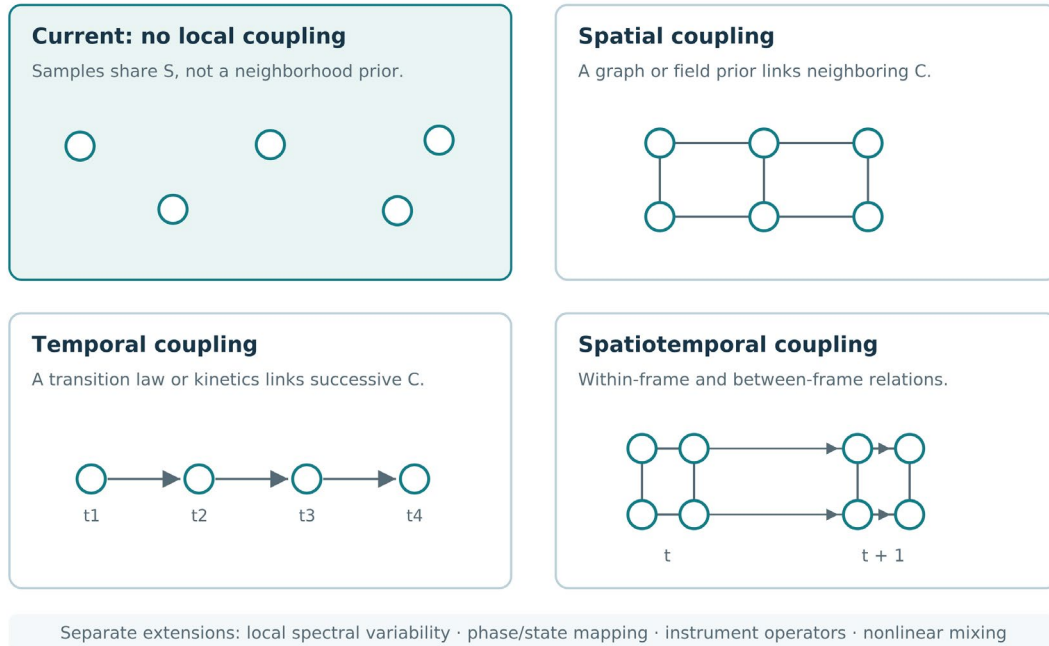


Figure 34. Extensions beyond the current static model. Spatial, temporal, and spatiotemporal priors couple different latent objects; they should be added only after specifying what is physically expected to vary locally.

9.4 Instrument Response and Counting Statistics

For counting instruments, the Gaussian residual model is only an approximation. The forward model should predict expected counts after efficiency, exposure, energy redistribution, calibration, and background, and the observation should follow an appropriate counting likelihood. A zero count is an ordinary observation, not a missing datum. At low dose, shot noise dominates the count covariance; at high dose, epistemic spectral and calibration uncertainty can dominate even though the conditional counter remains Poisson.

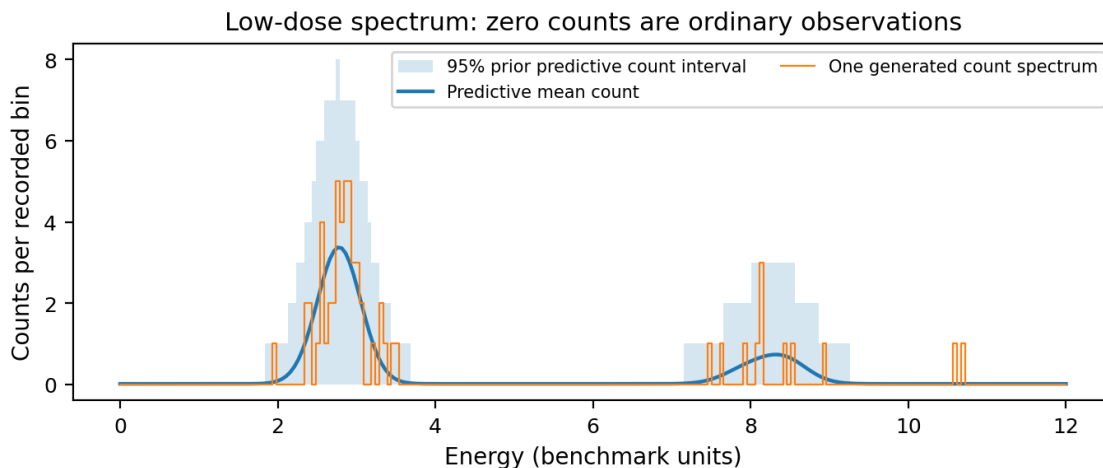


Figure 35. Low-dose count spectrum from the instrument/counting tutorial. Most channels are zero, and those zeros carry information about the latent rate.

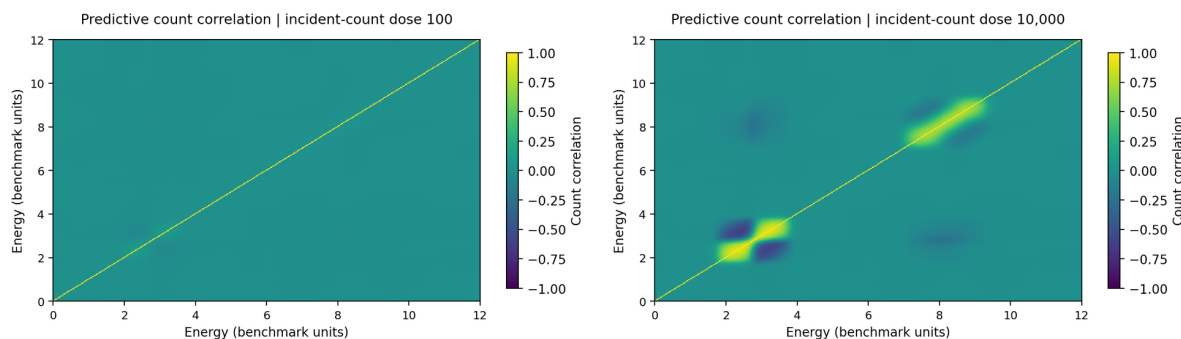


Figure 36. Count correlation at low and high dose for the same intrinsic generator. At low dose the shot-noise diagonal dominates; at high dose the off-diagonal contribution from uncertain latent rates becomes visible.

10. Practical Workflow for a New Materials Dataset

For a new dataset, start from the materials question rather than from the algorithm. Define the components, define what the reported fractions mean, establish how the spectra were measured, and decide which uncertainties are experimental and which represent incomplete prior knowledge. Only then choose Level 0, 1, or 2 and the numerical inference method. This order prevents a mathematically convenient factorization from silently changing the physical meaning of the result.

Step	Why it matters
1. Define the physical components.	State what one component means and whether its spectrum is globally shared.
2. Define the fraction convention.	Decide whether the weights are signal fractions or physical fractions; calibrate yields if necessary.
3. Separate the three uncertainty sources.	Measurement noise, composition uncertainty, and spectral uncertainty should be distinct model objects.
4. Validate composition geometry.	Use simplex-native coordinates or a raw assay model; do not invent independent Gaussian fraction errors after closure.
5. Choose Level 0, 1, or 2 from the form of spectral knowledge.	Do not choose a level because it sounds more advanced; choose it because it matches the provenance of the reference information.
6. Run prior-predictive checks.	The prior should be capable of generating measurements resembling the experiment.
7. Fit and inspect residual structure.	Systematic residual peaks or spectral autocorrelation usually indicate missing physics, a missing component, or a wrong noise model.
8. Report both accuracy and uncertainty diagnostics.	For synthetic or calibration data report RMSE and coverage. For real data report posterior width, sensitivity to prior assumptions, and held-out predictive checks.
9. Inspect covariance, not only marginal bands.	Covariance identifies which physical deformations remain uncertain and which variables compensate one another.
10. Escalate the model only when a failure mode requires it.	Add spatial coupling, composition-to-phase maps, local endmember variability, nonlinear mixing, or Poisson counting as explicit extensions—not as unexplained covariance inflation.

11. Main Lessons

Most materials unmixing problems are not blind factorizations of Y into W and S . In a real experiment we normally know something about the local composition, something about the expected component spectra, and something

about the measurement noise. None of these is exact. The useful problem is therefore to combine these pieces of information without pretending that one of them is known perfectly. Level 0 is the reference case with fixed spectra. Level 1 is appropriate when the spectral uncertainty can be described by a small number of physical parameters. Level 2 is appropriate when the natural prior information is a family of complete spectra. The progression should stop there unless the data require a more complicated model. Composition-to-phase conversion, solid-solution manifolds, spatial or temporal coupling, instrument response, and counting statistics are separate physical effects and should be added as separate model blocks, not absorbed into a generic uncertainty term. The central requirement is simple: every variable should retain a clear materials meaning, and every uncertainty should have a traceable experimental or physical origin.

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